MECHENG 706
Project 1 Information Brief

Title: Farm Tilling Service: Proof-of-Concept (Localization, Mapping & Path Following)

Team Size: 4 students

Due: Friday, Week 6 with live demo in lab

Project 1 Lead Instructor: Dr. Jaspreet Singh Dhupia

GTAs: Tamara Land, Hesam Hosseinizadeh, Siqi Jiang, Yitong Zhu

Walled Table Arena (Not to Scale)

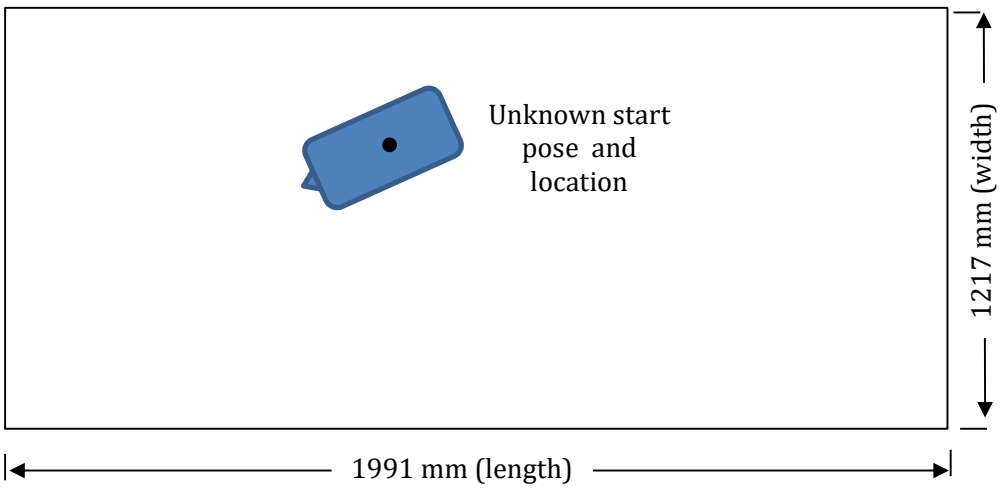

Figure 1. Walled Table Arena (not to scale). Dimensions approximate and for reference.

1. Challenge & Context

You are a team within AgriService Ltd. Develop and demonstrate a lab-scale automated tilling solution that can orient itself from an unknown initial pose, follow an efficient rectangular coverage path, and produce a usable mapping/coordinate output, using only the MECHENG 706 kit.

2. Kit, Constraints & Arena

Mobile base: Mecanum chassis with 4 motors (no encoders), Arduino Mega, motor drivers; do not modify the chassis or mobile robot hardware.

Sensors/actuators: IMU/MPU (use single-axis gyro), up to two long-range IR and two medium-range IR sensors, one ultrasonic sensor, one small servo; wireless module for logging; discrete components available; minor 3D-printed brackets allowed; no other major parts.

Arena: Fixed walled table; robot begins at an unknown location and orientation. Assessment is via an overhead camera tool measuring straight-line error, area covered, yaw deflection, and elapsed time.

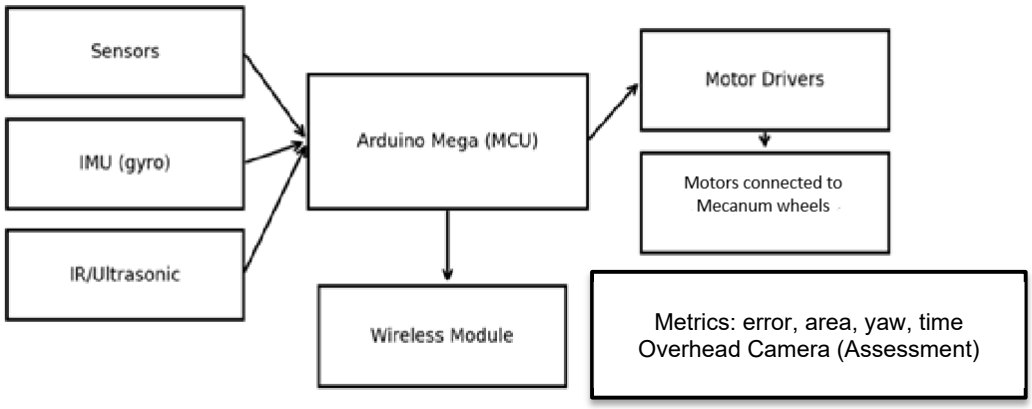

Figure 2. System architecture for Project 1.

3. Tasks & Execution

- Wiring and calibration of sensors; establish coordinate frames.

- Implement seek/align behavior using appropriate sensor fusion with range cues to find a boundary/corner.

- Implement path-following for rectangular coverage with appropriate speed and turn-rate constraints.

- Generate mapping/coordinates

4. Demo Protocol (Week 6)

1) Staff place the robot at an unknown pose. 2) Team starts the run. 3) Assessment software records metrics. 4) Official score is captured by TA. Base points may be awarded for genuine attempts that fail objectives.

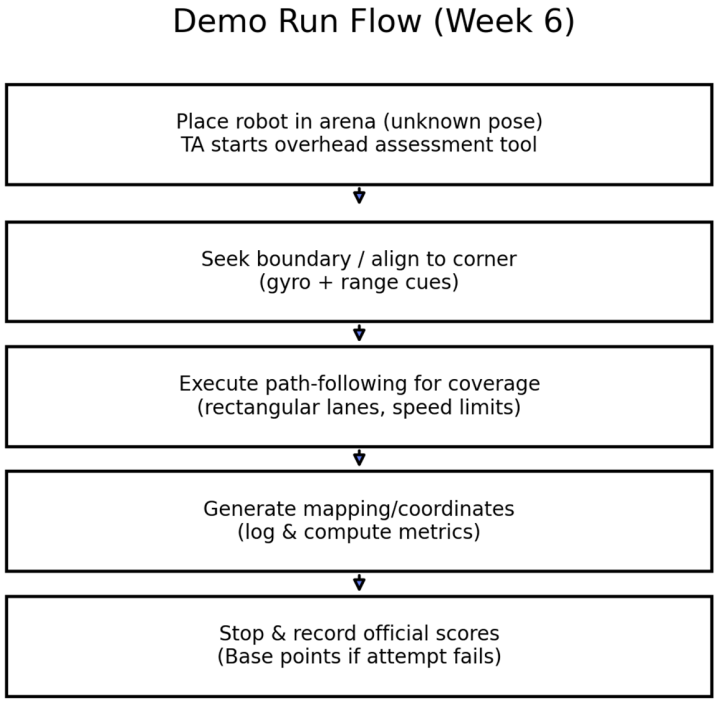

Figure 3. Demo run flow and scoring capture.

5. Deliverables & Submission

1. **Live demonstration under staff supervision** (assessment software is used to score your run).

2. **Results Summary (Individual submission collated for the group)**: 1 page per student on a DISTINCT technical aspect with original plots/figures. The result summary SHOULD NOT repeat content between group members.

- **Contract Proposal (Group, 4 pages):** Statement of Work, responsibilities and safety; itemised quote (≤ NZD $15,000 expected total); hazards & mitigations; references page for due diligence.

Submit two PDFs via Canvas by Friday of Week 6: (i) collated Results Summary (one per student), (ii) Contract Proposal.

6. Marking Rubric (100%)

Criterion	Bands / Description	Weight
Path Following — Completion	100% complete (10); 75–100% (8); 50–75% (6); Base (4)	10%
Path Following — Deviation	≤ ±10 mm (10); 12–14 mm (9); 14–16 mm (8); 16–18 mm (7); 18–20 mm (6); ≥ 20 mm (5); Base (4). Yaw > ±8° penalised.	20%
Time	≤ 90 s (10); 90–100 s (9); 100–110 s (8); 110–120 s (7); 120–130 s (6); ≥ 130 s (5); Base (4)	20%
Mapping & Coordinates	Generated map proportion: 100% (10); 75–100% (8); 50–75% (6); Base (4)	10%
Results Summary (Individual submissions collated as a Group)	Technical correctness & insight; original plots/figures; clarity & brevity; distinct contribution (20 pts total)	20%
Contract Proposal (Group)	SOW & responsibilities; safety; costed quote (≤ $15k); hazards & mitigations; professionalism & evidence (20 pts total)	20%

7. Safety, Inductions & Generative AI

- Complete MDLS online induction (quiz 10/10) and in-person induction before lab work; after-hours access requires declaration.

- AI tools can be used in the duration of the project for improving productivity in different tasks such as brainstorming, coding or proof-reading. Generative AI allowed (with disclosure) for Results Summary and Contract Proposal. In-person demonstration is mandatory as it follows the university's policy of including secure assessment for measuring learning outcomes.

- Ethical and responsible AI use is imperative and non-negotiable. Severe penalty of up to 100% of project score may be applied to individuals or groups for any identified cases of academic misconduct. Some examples of unethical and academic misconduct while using AI tools as

- Non-validated work components
- Reporting results that are not demonstrated
- Not proof-reading AI generated text
- Not validating / checking references. Including references from AI tools that are not directly relevant.